ALAKA ASRITHA GHANTASALA

AI & ML Enthusiast

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SKILLS

- **Languages**: Python, JavaScript, SQL
- **Web Technologies**: HTML5, CSS3
- **Machine Learning**: Scikit-learn, TensorFlow, Keras,Pandas,numpy
- **Tools**: Git, GoogleColab, Anaconda, Visual Studio,Streamlit
- **Databases**: MySQL
- **Data Structures & Algorithms**

EDUCATION

B.Tech, Computer Science, 2022- 2026
Specialization : AI & ML Woxsen University, India CGPA: 3.6/4

Intermediate, MPC, 2020-2022
Aditya Junior College, India
PERCENTAGE: 95%

Schooling , 2010-2020
Vidyanjali CBSE School, India
PERCENTAGE: 93.8%

PROFESSIONAL SUMMARY

Result-oriented and diligent individual with a proven ability to manage tasks effectively, prioritize well, and consistently meet deadlines. Known for a positive attitude, quick learning, and strong analytical thinking to support informed decisions. Committed to delivering high-quality outcomes through innovative problem-solving, clear communication, and effective collaboration to achieve goals.

EXPERIENCE

Machine Learning I ntern | National Institute od Rural Development Panchayati Raj (NIRDPR)
April2025 - July 2025

- Developed a deep learning model using MobileNetV2 to classify road conditions (Good, Bad, Potholes) from drone imagery, achieving ~90% accuracy through advanced data augmentation.
- Preprocessed large-scale geospatial datasets and segmented road regions from drone-captured tiles; manually labeled data to build a high-quality training set.
- Proposed future enhancements including severity-level classification, real-time road damage detection, and accident-prone zone prediction to aid smart city planning and public safety.

KEY PROJECTS

VitaminDeficiency Detection
Tech Stack: Python, TensorFlow/Keras, OpenCV, NumPy, Matplotlib, Scikit-learn.
July 2023- December 2023

- Designed a CNN model based on AlexNet to classify tissue images into five vitamin deficiency categories, achieving 85–95% accuracy for potential clinical diagnostics.
- Engineered image preprocessing (resizing, normalization, segmentation), boosting training efficiency by 18%.
- Proposed and optimized a mobile deployment framework using Softmax-based multiclass predictions, enabling real- time diagnosis in remote healthcare environments.

SHESAFE(ANDROIDAPP)
Tech Stack: Python, TensorFlow/Keras, OpenCV, NumPy, Matplotlib, Scikit-learn.
January 2024- April 2024

- Trained an LSTM model on MFCC features from the RAVDESS dataset to detect speech-based stress, achieving 91% accuracy in real-time emotional monitoring.
- Developed an Autoencoder and applied One-Class SVM on MobiAct data to detect motion anomalies achieving a 0.87 F1- score.
- Embedded models with TensorFlow Lite for offline SOS alerts, cutting response time by 3 mins, and aided real-time mobile UI/UX.

CERTIFICATIONS

- Introduction to Programming- Kaggle
- Introduction to tensorflow for AI,ML,DL - Coursera
- Introduction to Deeplearning and Neural Networks Keras - Coursera

INTERESTS

- Reading Books
- Exploring new Tech
- Athletic Activities

LANGUAGES

- English
- Telugu
- Hindi

BHASHINI LLM : Hindi - English Translation with RAG
Tech Stack: Python, FastAPI, FAISS, Sentence Transformers, Hugging Face Transformers, RAG, NumPy.
December 2024 - February 2025

- Devised a multilingual translation engine by integrating FAISS-based semantic search with MarianMT models for context-aware Hindi-English translation. Implemented Sentence
- Transformers for vector embeddings, enhanced translation accuracy by 28% using Retrieval-Augmented Generation (RAG).
- Deployed a FastAPI production backend with Whisper API for speech-to-text, reducing latency to under 1.5 seconds, and scaled support to 5+ regional languages with 90%+ BLEU score consistency.

NavTeach : Teacher Navigation System Tech Stack: Embedded C/C++, Arduino, RFID, Zigbee, XBee, Python, Serial Communication, Matplotlib (for visualization), TinkerCAD (for prototyping). January 2025 - November 2025

- Engineered a campus-wide teacher tracking system using RFID and Zigbee modules, enabling accurate indoor localization across 50,000+ sq. ft. without GPS dependency.
- Integrated low-power, long-range Zigbee sensor networks, eliminating reliance on mobile data or internet for seamless indoor communication.
- Enhanced tracking efficiency by 30% over GPS/manual methods through real-time visualization and customized digital campus layouts.

ACHIEVEMENTS

- Winner, SAWiT AI Hackathon 2025 – Recognized among Top 25 Winning Teams.
- Awarded Silver Rank (Top 2%) among 500+ peers for outstanding academic excellence, 2024.